

LING 411 Problem Session: Hypothesis Formation

Answer Key

4. Questions

- Q1.** What is the research question of the paper? The paper names two hypotheses (Hypothesis 1 and Hypothesis 2). State each in your own words, and say which one is H (the one the authors argue for) and which one is H'.

Answer. Question: when learners infer the word order of a new language from incomplete evidence, do they use abstract structure (scope) or surface statistics (which words follow which in English)? **Hypothesis 1:** the new language is English order with the noun moved to the front. **Hypothesis 2:** the new language orders modifiers by scope, so it is the mirror image of English. The authors argue for Hypothesis 2, so Hypothesis 2 = H and Hypothesis 1 = H'. (Note that their numbering is the reverse of what you might expect.)

- Q2.** Fill in the table: which order does each hypothesis predict at test?

Training → Test	H' (surface)	H (scope)
N-Adj, N-Dem → {N, Adj, Dem}	N-Dem-Adj	N-Adj-Dem
N-Num, N-Dem → {N, Num, Dem}	N-Dem-Num	N-Num-Dem
N-Adj, N-Num → {N, Adj, Num}	N-Num-Adj	N-Adj-Num

Answer. The two hypotheses make *opposite* predictions in every condition. H' predicts N-Dem-Adj because the English chunk *this green* (Dem-Adj) is kept. This is what lets the experiment decide between H and H'.

- Q3.** Why did each training phrase contain only *one* modifier? What would go wrong if learners had heard phrases like *vase purple those*?

Answer. If learners heard two modifiers together, the training would show them the correct order directly. They would simply copy what they heard, so their choices would tell us nothing about their own preferences. Hiding this evidence is what makes the two hypotheses give different predictions.

- Q4.** Why did the authors put the modifiers *after* the noun? Would the experiment work if they came *before* the noun, as in English?

Answer. No. Before the noun, both hypotheses predict the same order: English order is Dem-Adj-N, and the scope order is also Dem-Adj-N. The predictions only come apart when the noun is at the other edge. A good test needs a case where H and H' predict different data.

- Q5.** Which hypothesis do the results support? Is H' fully ruled out?

Answer. They support H: in all three conditions most choices followed the scope order (Adj/Dem 0.94, Num/Dem 0.70, Adj/Num 0.75). H' is not supported, but it is not fully ruled out: in two conditions about 25–30% of choices still followed the surface order. The fair conclusion is that the scope preference is *stronger* than the surface preference, not that surface statistics play no role.

- Q6. Think further.** Propose another alternative, H'', that could also explain the results without scope. How could we test it?

Answer. One possible H'': a simple *mirror* strategy: “if the noun moves to the other side, reverse everything.” Reversing English Dem-Adj-N gives N-Adj-Dem, the same as the scope prediction. With English speakers the two cannot be separated, because English order is already scope-isomorphic. A follow-up could test speakers of a language whose order is *not* scope-isomorphic (e.g., N-Dem-Num-Adj) on a language with modifiers before the noun: mirroring predicts Adj-Num-Dem-N, but scope predicts Dem-Num-Adj-N. (Other good answers are possible.)